Suggested problems: 1-19 odd

Problem 1: Give cartesian coordinates for the following points with given polar coordinates:

(a) $(5, 0)$ $(5, 0)$

(b) $(5, \pi)$ $(-5, 0)$

(c) $(1, 2\pi/3)$ $(-1/2, \sqrt{3}/2)$

(d) $(2, 5\pi/6)$ $(-\sqrt{3}, 1)$

Problem 2: Give polar coordinates for the following points with given cartesian coordinates:

(a) $(1, 1)$ $(\sqrt{2}, \pi/4)$

(b) $(-1, 0)$ $(1, \pi)$

(c) $(\sqrt{6}, -\sqrt{2})$ $(\sqrt{8}, -\pi/6)$

(d) $(-\sqrt{3}, 1)$ $(2, 5\pi/6)$

Problem 3: Graph the following functions in the xy-plane.

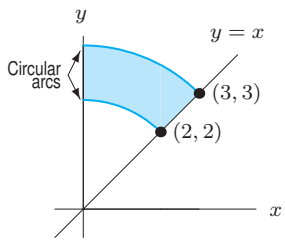
(a) $r = 1$ See class notes.

(b) $r = \theta$ See class notes.

(c) $r = 1 - \sin(\theta)$ See class notes.

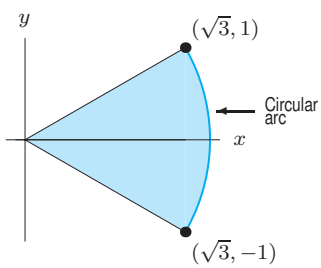
Problem 4: Give inequalities in terms of r and θ which describe the following regions in the xy-plane.

(a)



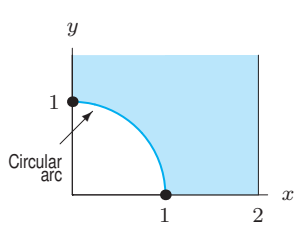
$\sqrt{8} \leq r \leq \sqrt{18}$, $\pi/4 \leq \theta \leq \pi/2$

(b)



$0 \leq r \leq 2$, $-\pi/6 \leq \theta \leq \pi/6$

(c)



Note: Region extends indefinitely in the y-direction.

$1 \leq r \leq \frac{2}{\cos\theta}$, $0 \leq \theta \leq \pi/2$